## Clio API Orchestratir - Web App Mock Up

Quill & Arrow Logo + Home Button
Clio Data Management Tool

Settings | Exit Button

**Page Title - Matter, Custom Field, Contacts Operations, or Audit Log**

Internal/Current Main Page Navigation Tabs Area

Button | Button | Button | Button

- Side Navigation Menu Option
- Side Navigation Menu Option
- Side Navigation Menu Option
- Side Navigation Menu Option

Data Information Boxes - Populated with Metrics of Operations

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**Report on Bulk Updates, Lists, or Export/Import Results Title**

**List View Data Fields (Dynamic)**

| matter_id | description | status | location | client_reference | open_date | closed_date | pending_date | billable |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
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| | | | | | | | | |

**Mini Audit Log**

- List showing the most recent operators that have been executed and by which user
- Should be very basic information, but visually appealing

## Clio API Orchestrator - Web Application Front & Back End Breakdown

### Target Web App Architechure

**Azure Web App**

**React SPA (Single Page Application) - Frontend**
- Login
- Dashboard
- Bulk Operations
- Audit Log
- Settings

→

**FastAPI Backend**
/api/auth → login
/api/matters → list/search
/api/preview → dry run
/api/execute → real run
/api/audit → log viewer

**Core Engine**
clio_client.py + operations.py

↓

**Clio API v4**

**SQLite - Audit Log - Upgrade to Azure SQL Later**

### Front End - File/Folder Map

```
src/
├── main.jsx        ← Entry point. Just mounts <App /> into the page. You'll never touch this.
├── App.jsx         ← The "router." Maps URL paths to page components.
├── index.css       ← Where Tailwind is imported. Global styles go here.
└── App.css         ← Mostly unused now that we use Tailwind.

├── api/
│   └── client.js           ← Every API call goes through here. Handles auth tokens automatically.

├── context/
│   └── AuthContext.jsx     ← Stores "who is logged in." Provides login/logout to the whole app.

├── components/
│   └── Layout.jsx          ← The sidebar + top header shell. Every page renders inside it.

├── pages/
│   ├── LoginPage.jsx       ← The login screen (standalone, no sidebar)
│   ├── DashboardPage.jsx   ← The landing page after login (KPI cards + activity feed)
│   ├── MattersPage.jsx     ← Matter listing table
│   ├── CustomFieldsPage.jsx ← Custom fields listing table
│   ├── BulkOperationsPage.jsx ← The 3-tab page we just built
│   └── AuditLogPage.jsx    ← Full audit history table

├── assets/         ← Images (hero.png, SVGs). Only referenced, never logic.
```

### 3 Layer Universal Architecture

**Layer 1: Layout.jsx**

**The Fram Around Every Page**

**This is the sidebar + top header. When you want to change:**

- Sidebar navigation labels → find the NAV_ITEMS array near the top (line 23). Each entry is { to: '/url', icon: IconName, label: 'Visible Text' }.
- Sidebar colors → the <aside> uses bg-slate-900 (dark background). Active nav item is bg-blue-600. Inactive text is text-slate-300.
- Brand name → lines 53-54: "Quill & Arrow" and "Clio Data Management".
- Top header → the <header> tag around line 96. Search bar placeholder, user name display, logout button are all here.

**Layer 2: Each Page File**

**What Appears in the Main Content Area**

**Every Page Follows the Exact Same Skeleton:**

```jsx
export default function SomePage() {
  // state variables at the top
  // useEffect to fetch data on load
  // return JSX
  return (
    <div>
      <h1 className="text-2xl font-bold text-slate-800 mb-6">Page Title</h1>
      {/* cards, tables, forms below */}
    </div>
  );
}
```

- To change a page title, find the <h1> — it's always the first element.
- To change a card's appearance, look for the pattern bg-white rounded-xl shadow-sm border border-slate-200 — that's the standard card style used everywhere.
- To change a table's columns, find the <thead> and its <th> elements for headers, then the matching <td> elements in <tbody>.

**Layer 3: API/Client.js**

**How Data Get Fetched**

**Three functions are used throughout:**

- get('/some-endpoint') — fetches data
- post('/some-endpoint', { key: value }) — sends JSON
- postForm('/some-endpoint', formData) — sends file uploads

You won't need to touch this for visual changes.

### Advanced Search & Detail Drill Down

#### Backend

**operations.py** → **5 new functions added**

- **get_matter_detail(client, matter_id)** — fetches a single matter with ALL 25+ fields (identity, dates, people, practice area, financials, stage, group, relationships, custom field values with resolved names)
- **search_matters(client, query, limit)** — searches/lists matters with people and practice area included in results (lightweight for table display)
- **get_custom_field_detail(client, field_id)** — fetches full custom field metadata (required, displayed, deleted, picklist_options with option text, custom_field_set)
- **search_custom_fields(client, query, parent_type, limit)** — searches fields by name with optional parent_type filter
- Constants **MATTER_DETAIL_FIELDS, MATTER_SEARCH_FIELDS, CUSTOM_FIELD_DETAIL_FIELDS** — define exactly which fields to request from Clio

**backend/routes/matters.py** → **rebuilt with:**

- **GET /api/matters** — unchanged basic listing
- **GET /api/matters/search** — new advanced search with 8 filter parameters: q, responsible_attorney, originating_attorney, responsible_staff, open_date_from, open_date_to, status, cf_filters (JSON array for custom field value filtering)
- **GET /api/matters/{matter_id}** — now returns full enriched detail

**backend/routes/custom_fields.py** → **rebuilt with:**

- **GET /api/custom-fields** — unchanged basic listing
- **GET /api/custom-fields/search** — new search with q, field_type, parent_type filters
- **GET /api/custom-fields/{field_id}** — new full detail endpoint

#### Frontend

**MattersPage.jsx**

- **Row 1:** Matter ID/Display Number (text + search icon), Open Date From (date picker), Responsible Attorney, Search button
- **Row 2 (toggleable "Advanced Search"):** Originating Attorney, Responsible Staff, Custom: O/C, Custom: Manufacturer, Custom: Trial Date
- *"Clear filters"* link when any filter is active
- Results table with 7 columns (chevron, display number, description, status, open date, resp attorney, practice area)
- **Click any row to expand inline** → fetches full detail from /api/matters/{id} and renders a 3-column panel with all fields organized into Identity & Status, Dates & Details, People & Financials sections, plus custom field values table and relationships list

**backend/routes/matters.py**

- **Search:** Field ID/Name (text), Field Type (dropdown with all Clio types), Parent Type (dropdown: All/Matter/Contact/Activity)
- Results table with 7 columns (chevron, ID, name, type badge, parent, required, displayed)
- **Click to expand** → full detail panel with Identity, Configuration (type, required, displayed, deleted badges), Field Set info, and Picklist Options table

**Layout.jsx** → **search bar placeholder changed from "Search matters..." to "Search..."**

### React JS + Tailwind CSS Code Operation Guide

**Tailwind CSS =** CSS framework designed to speed up web development by allowing developers to style elements directly in their HTML markup

**React + Tailwind CSS:** commonly paired to build modern web applications using a "utility-first" approach. Instead of writing custom CSS files, you style React components by applying pre-defined utility classes directly to elements.

**Every color/size/spacing in this project is a Tailwind utility class**

| Element Being Updated | Class to look for | Example alternatives |
|---|---|---|
| | | |
| Text color | text-slate-800 | text-slate-600 (lighter), text-blue-600 (blue) |
| Background | bg-white, bg-slate-50 | bg-blue-50 (light blue tint) |
| Font size | text-2xl, text-sm, text-xs | text-lg, text-base, text-3xl |
| Font weight | font-bold, font-semibold, font-medium | Just swap between them |
| Spacing | mb-6, p-5, gap-4 | The number = units of 0.25rem (4=1rem, 6=1.5rem) |
| Rounded corners | rounded-xl | rounded-lg (less), rounded-2xl (more) |
| Border | border border-slate-200 | border-slate-300 (darker line) |
| Status pill (green) | bg-green-100 text-green-700 | Swap green for any color |
| Status pill (red) | bg-red-100 text-red-700 | Same pattern |

**Quick Reference for Changes/Updates**

| Change Action | File Location | Element Key |
|---|---|---|
| | | |
| Change the sidebar brand name | Layout.jsx | "Quill & Arrow" |
| Change sidebar nav labels | Layout.jsx | NAV_ITEMS array |
| Change sidebar colors | Layout.jsx | bg-slate-900, bg-blue-600 |
| Change a page title | The page's .jsx file | The <h1> tag |
| Change table columns | The page's .jsx file | <th> and <td> elements |
| Change the login page branding | LoginPage.jsx | "Quill & Arrow", the bg-blue-600 icon box |
| Change button colors | Whichever page has the button | bg-blue-600 (primary), bg-emerald-600 (execute) |
| Add a new page | App.jsx | Add a <Route> inside the protected block, create a new file in pages/ |
| Add a new sidebar link | Layout.jsx | Add an entry to NAV_ITEMS |